

GEOGRAPHY

PHYSICAL GEOGRAPHY

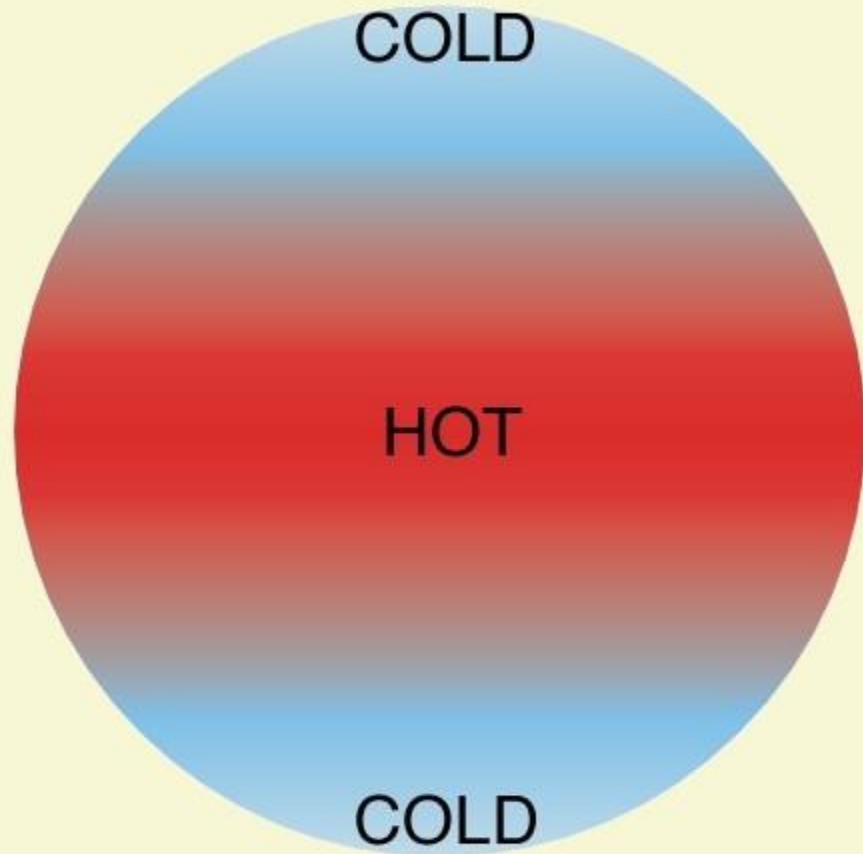


CLIMATOLOGY

MODULE – 18

- GLOBAL PRESSURE BELTS
- PLANETARY WIND DISTRIBUTION

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE



No regular distribution of pressure because pressure is NOT strictly temperature dependent

- 1 There is **no definite trend of distribution** of pressure from equator towards the poles.
- 2 If the air pressure would have been the function of air temperature alone there should have been regular increase of **pressure poleward** because temperature regularly decrease from the equator towards the poles.
- 3 But **this is not the case**. This implies that there are factors more than temperature that control pressure distribution.

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

PRESSURE BELTS ON THE BASIS OF MODE OF GENESIS

THERMALLY INDUCED PRESSURE BELTS

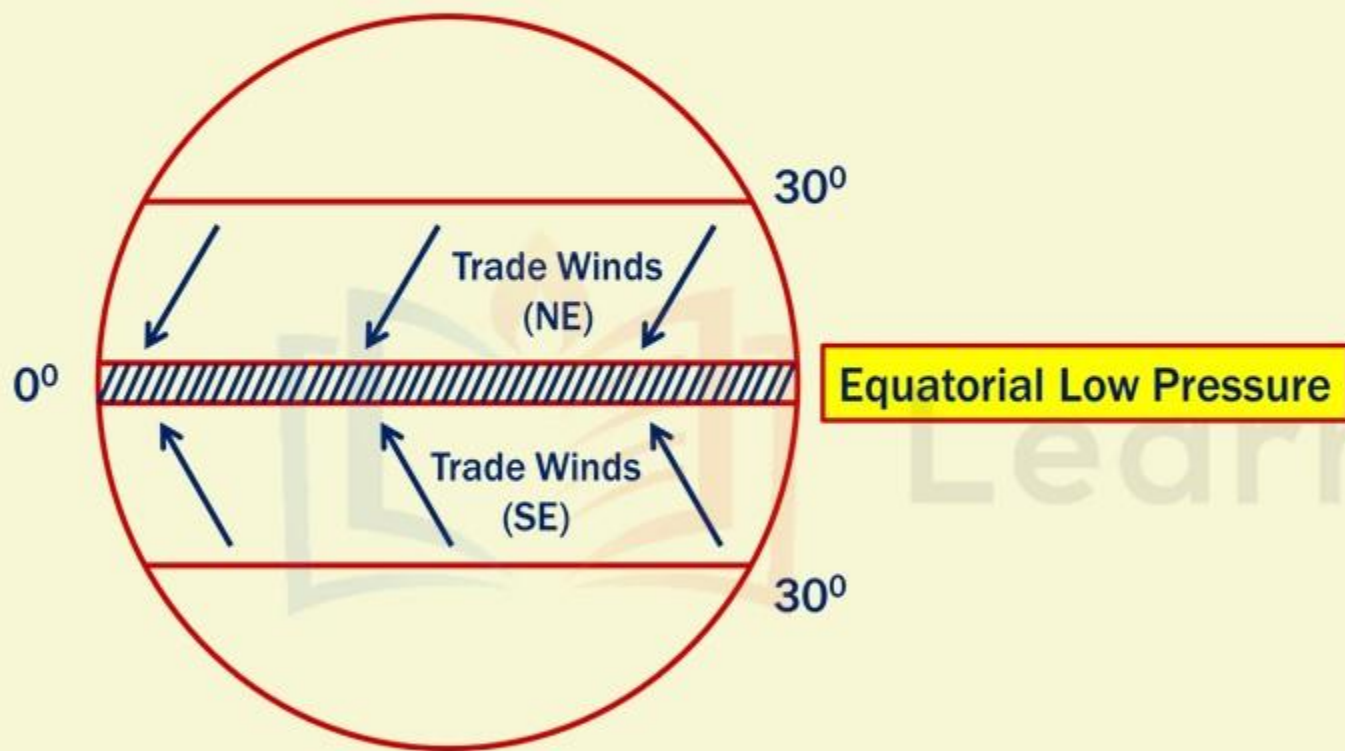
- 1** This is due to the influence of temperature by which the pressure tends to rise from the equator towards the poles.
- 2** Belts produced because of this:
 1. Equatorial low pressure
 2. Polar high pressure

DYNAMICALLY INDUCED PRESSURE BELTS

- 1** This is caused by the rotation of the earth by which air is pulled pulled inwards in the polar region. This causes a rarification of the atmosphere in the sub-polar latitudes.
- 2** Belts produced because of this:
 1. Sub-Tropical High Pressure Belt
 2. Sub-Polar Low Pressure Belt

In all, there are seven pressure belts on the globe.

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

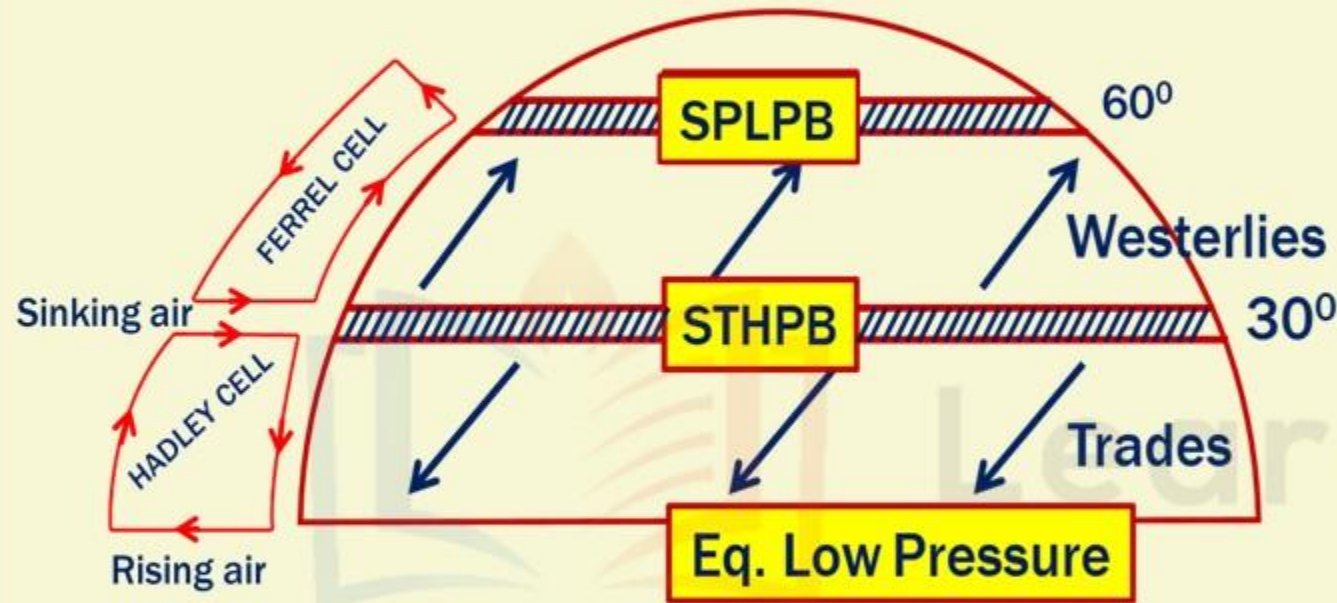


1.EQUATORIAL LOW PRESSURE BELT

- 1 It is located on either side of the geographical equator in a zone extending between **5°N-5°S latitudes**.
- 2 This belt is **thermally induced** because the ground surface is intensely heated due to almost vertical rays of the sun.
- 3 During the northern summer, this belt shifts up to 20° N in Africa and during southern summer, it shifts to 10° to 20°S latitude.(Reason-Continentality nature of Northern hemisphere)

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

2.SUB-TROPICAL HIGH PRESSURE BELT



1 Sub-Tropical high pressure belt extends between the **latitudes of 25°-35°** in both the hemispheres.

2 This high pressure belt is **not thermally induced** because it receives fairly high temperature throughout the year.

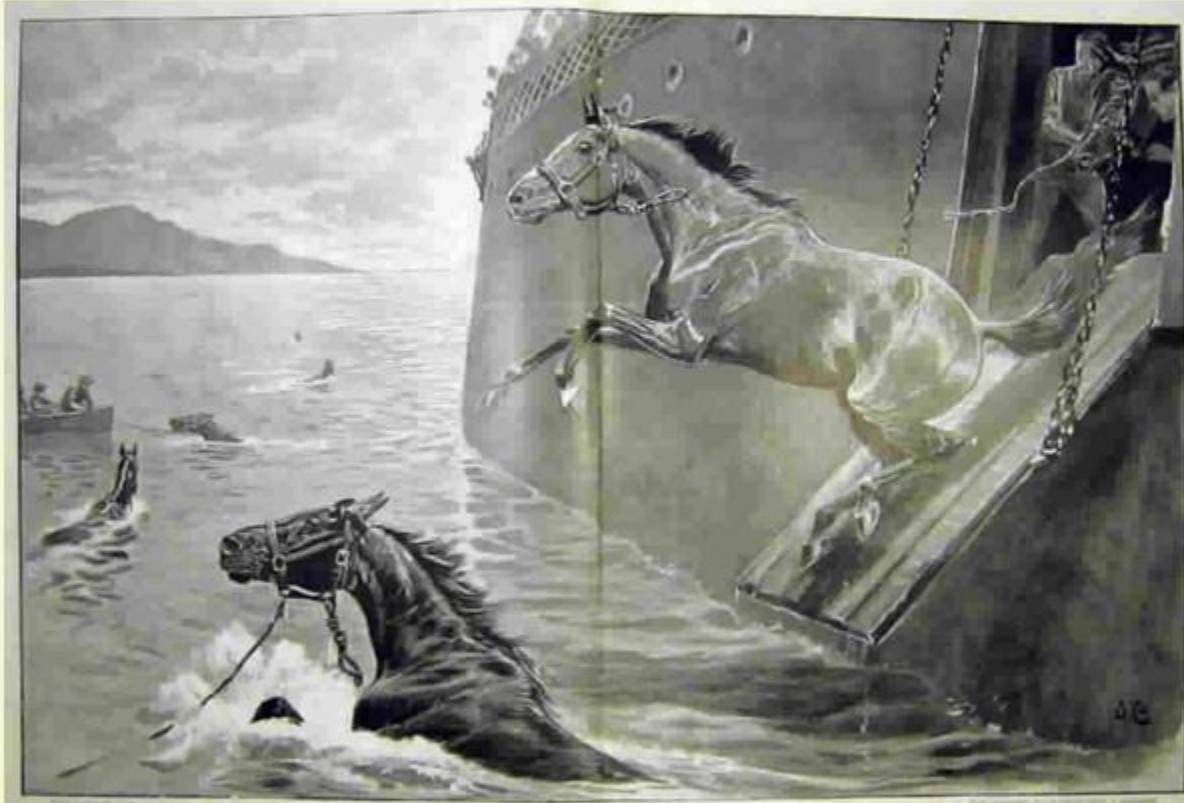
3 Thus this belt owes its **origin to the rotation of the Earth and sinking and settling down of winds**. It is, thus, apparent that the sub-tropical high pressure belt is **dynamically induced**.

4 The convergence of winds at higher altitude above this zone results in the subsidence of air from higher altitudes.

5 Thus, descent of winds results in the **contraction of their volume and ultimately causes high pressure**.

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

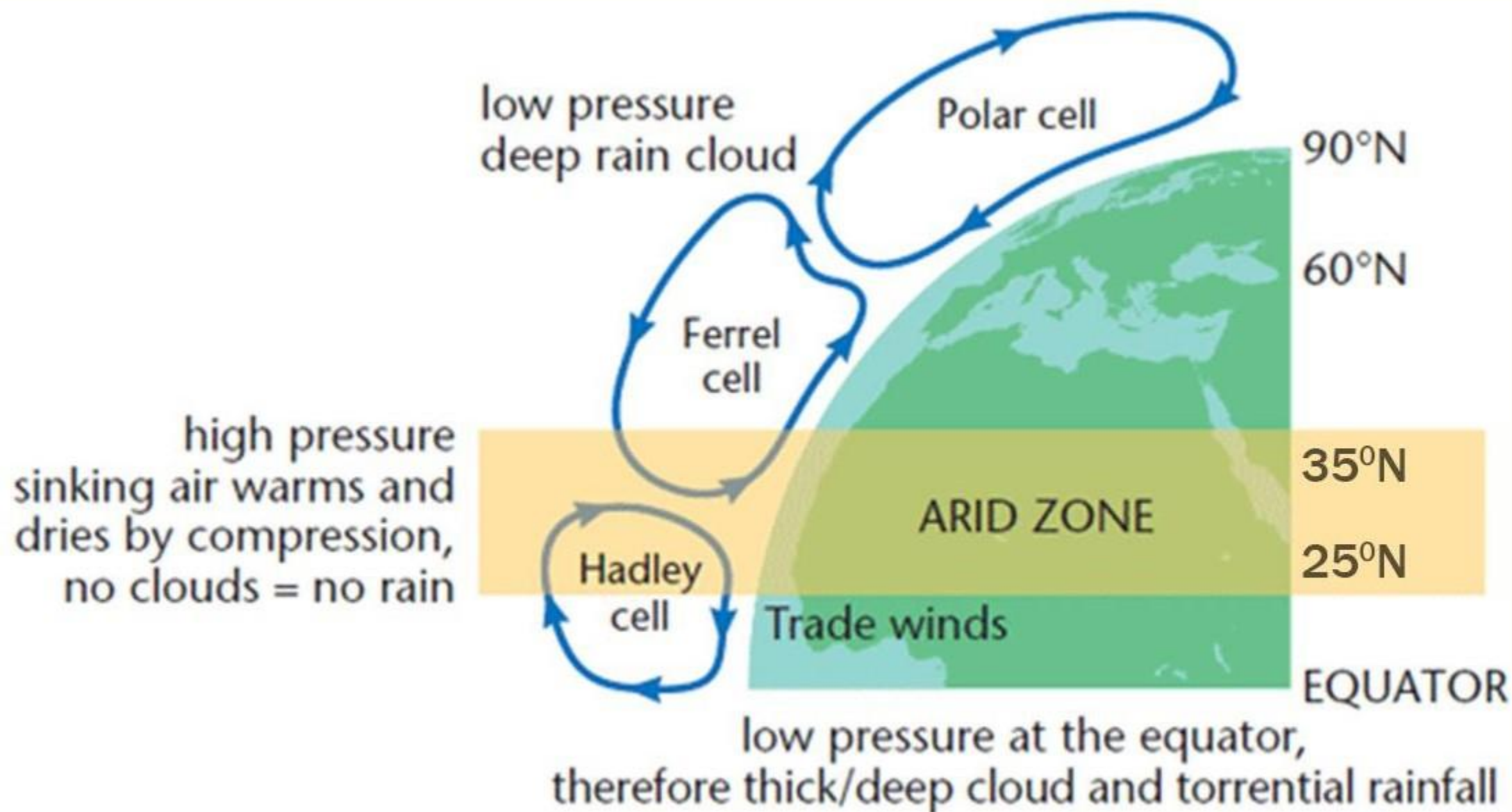
2.SUB-TROPICAL HIGH PRESSURE BELT



FUN FACT!!

Do you know why it is called “horse latitudes”?

- 6** This is why this zone is characterized by **anticyclonic conditions** which causes stability & aridity.
- 7** This is one of the reasons for the presence of hot deserts of the world in the western parts of continents in a zone extending between 25° - 35° in both the hemispheres.
- 8** This zone of high pressure is called ‘**Horse latitude**’ because of **prevalence of frequent calms**.
- 9** In ancient times, the merchants carrying horses in their ships, had to throw out some of the horses while passing through this zone of calm in order to lighten their ships.



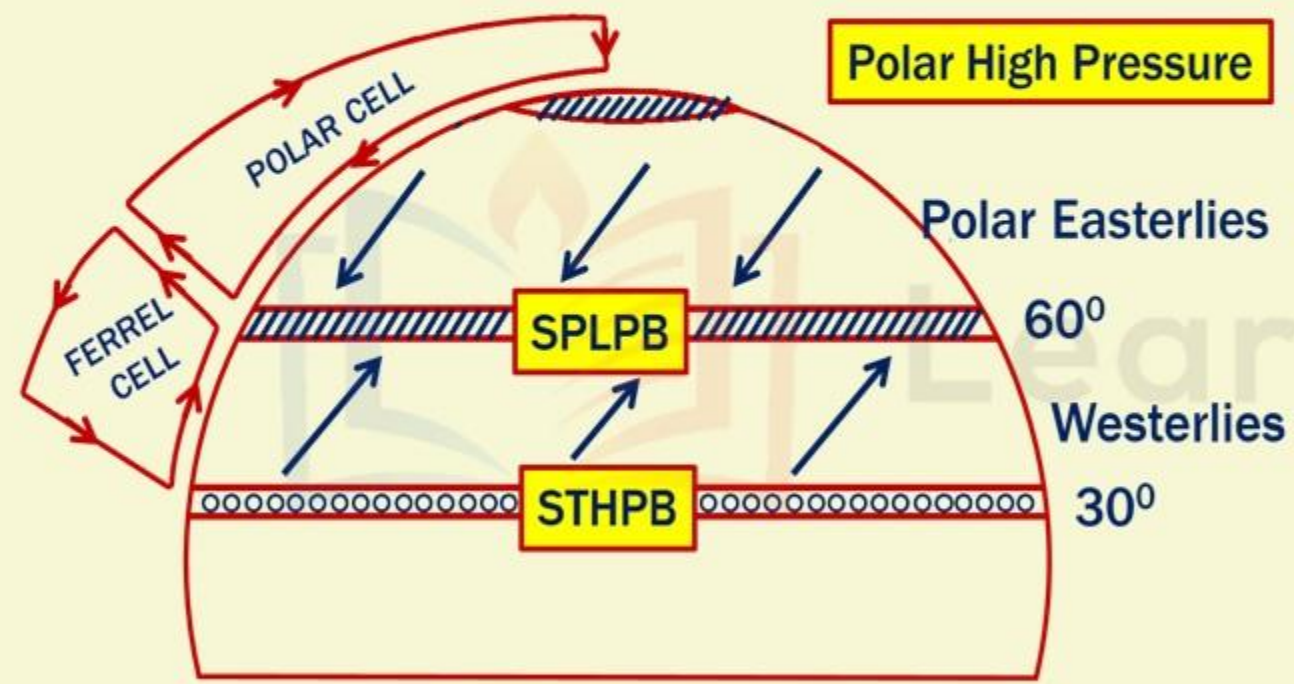
GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

3.SUB-POLAR LOW PRESSURE BELT

- 1 This belt of sub-polar low pressure is located between 60° - 65° latitudes in both the hemispheres.
- 2 The low pressure belt **does not appear** to be **thermally induced because there is low temperature** throughout the year and as such there should have been high pressure belt instead of low pressure belt.
- 3 It is thus obvious that **this low pressure belt is dynamically produced.**
- 4 In fact, the **surface air spreads outward from this zone due to rotation** of the Earth and low pressure is caused.
- 5 It may be pointed out that this factor should be more effective at the poles, but **effects of the rotation is negated or say overshadowed by exceptionally low temperature** prevailing throughout the year at the poles.

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

4. POLAR HIGH PRESSURE BELT

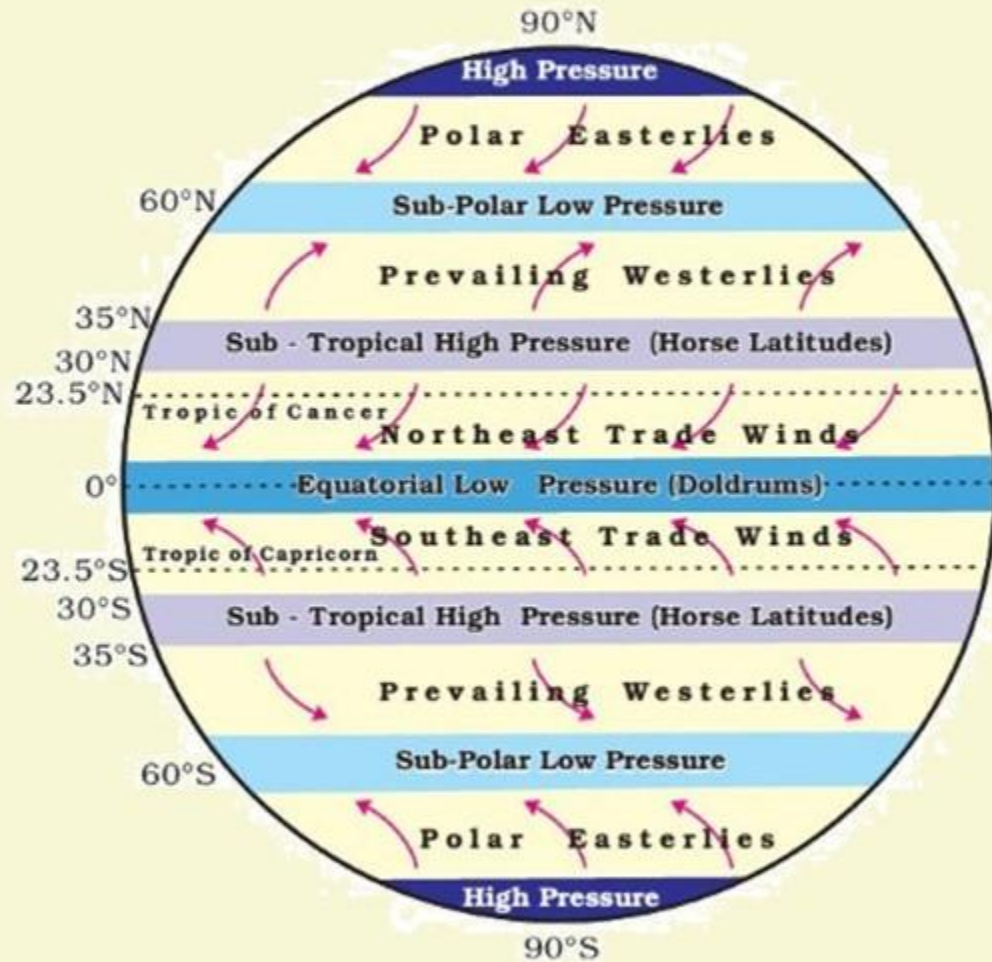


1 High pressure persists at the poles throughout the year because of prevalence of very low temperature (below freezing point) all the year round.

2 In fact, both the factors, thermal and dynamic, operate at the poles but the exceptionally low temperature or thermal factors present overpowers the dynamic factors.

GLOBAL PLANETARY DISTRIBUTION OF AIR PRESSURE

PERMANENT OR PLANETARY WINDS

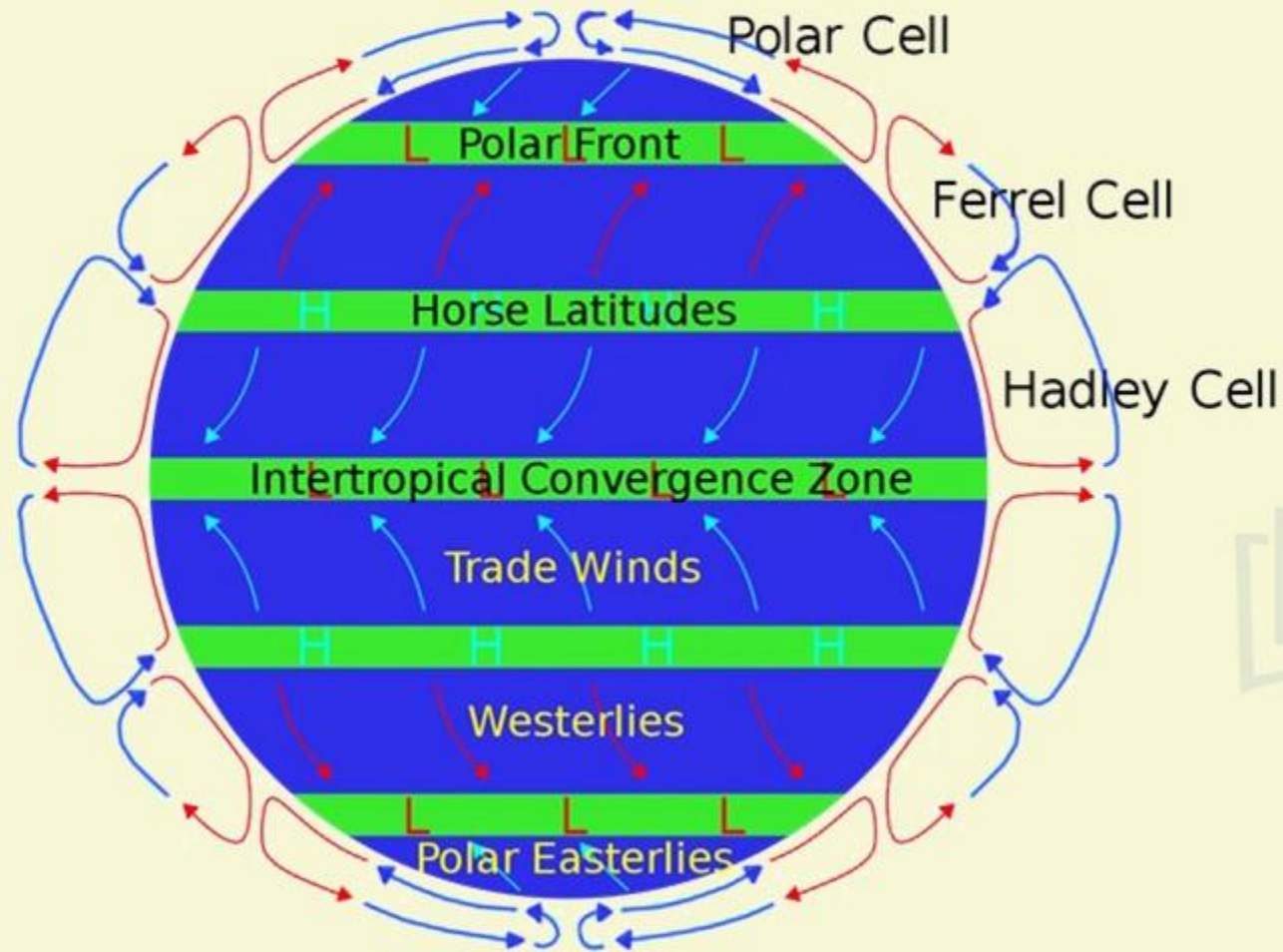


Major Pressure Belts and Wind System

- 1 On an average, the location of high and low pressure belts is considered to be stationary on the globe.
- 2 Consequently, winds blow from high pressure belts to low pressure belts
- 3 Since, these winds are distributed all over the world and these are related to thermally/dynamically induced pressure belts and rotation of Earth, they are called planetary winds.
- 4 These winds include trade winds, westerlies and polar winds.

PERMANENT OR PLANETARY WINDS

DOLDRUM AND EQUITORIAL WESTERLIES

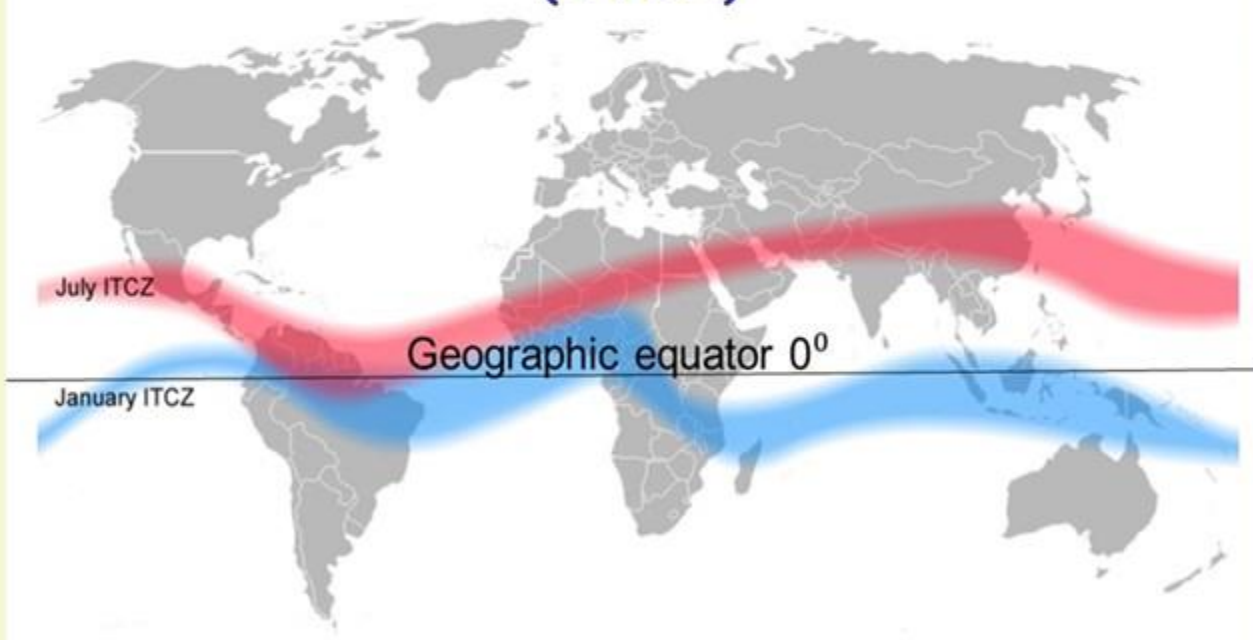


- 1** A **belt of low pressure**, popularly known as equatorial trough of low pressure or doldrum, extends along the equator within a **zone of 5° N and 5° S** latitudes.
- 2** This belt shifts northward during the summer solstice and then shifts southward at the time of winter solstice.
- 3** The equatorial or tropical fronts are called **intertropical fronts (ITF)** or **intertropical convergence (ITC)**.
- 4** This zone representing the meeting ground of north-east and south-east trade winds is known as ITCZ.

PERMANENT OR PLANETARY WINDS

DOLDRUM AND EQUITORIAL WESTERLIES

Earth's Thermal Equator, a.k.a., the INTERTROPICAL CONVERGENCE ZONE (ITCZ)



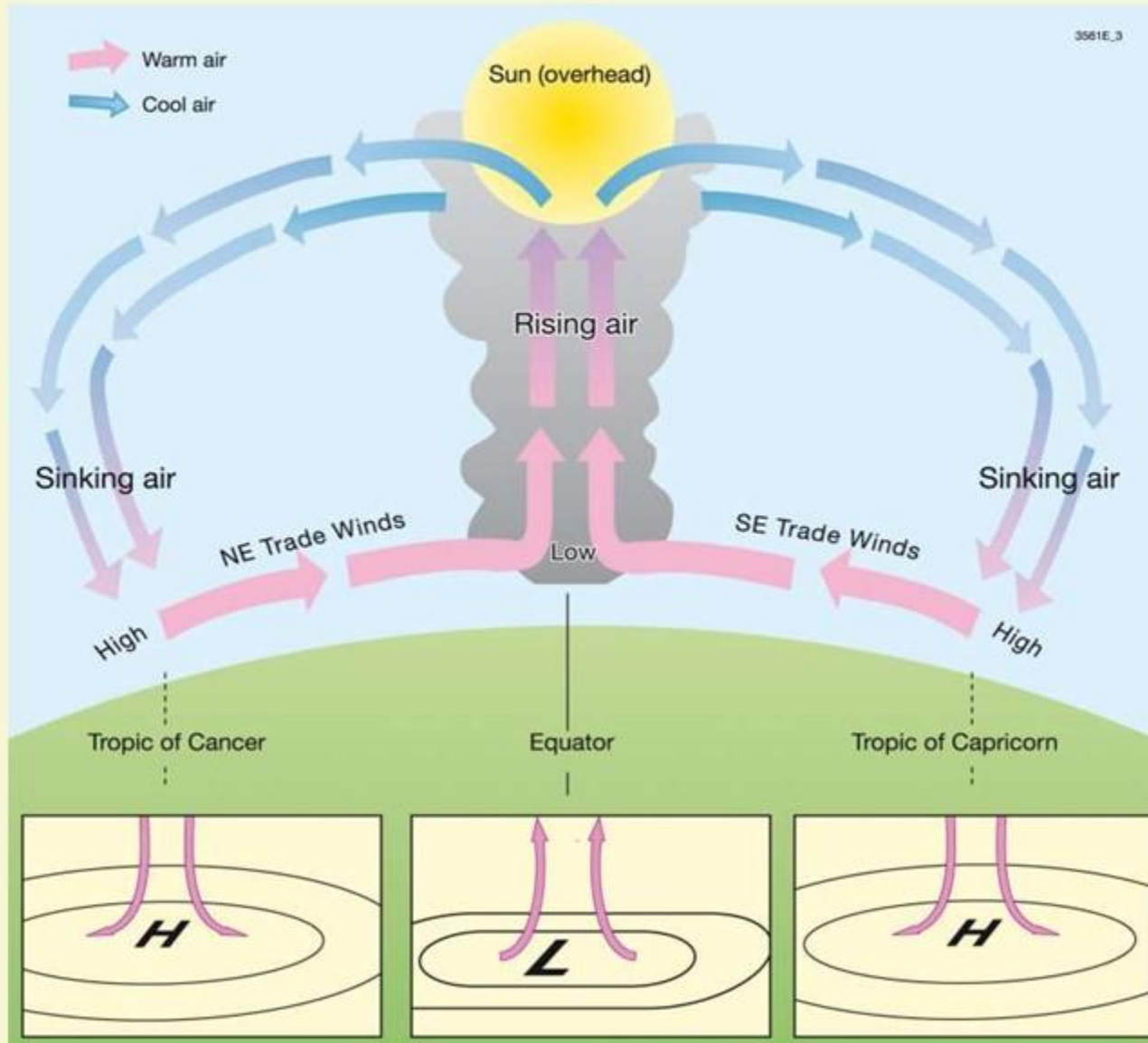
5 The northern and southern boundaries of intertropical convergence are called north intertropical convergence (NITC) and south intertropical convergence (SITC) respectively.

6 Because of the **seasonal shifting of the ITCZ**, during the northern summers (say for example) the SE trades will have to cross the equator to converge at the ITCZ.

7 While crossing the equator and moving to the northern hemisphere, **the winds will experience coriolis force pushing them towards right** and thus what was once the SE trades will attain a westerly direction and **become equatorial westerlies** in the doldrums.

PERMANENT OR PLANETARY WINDS

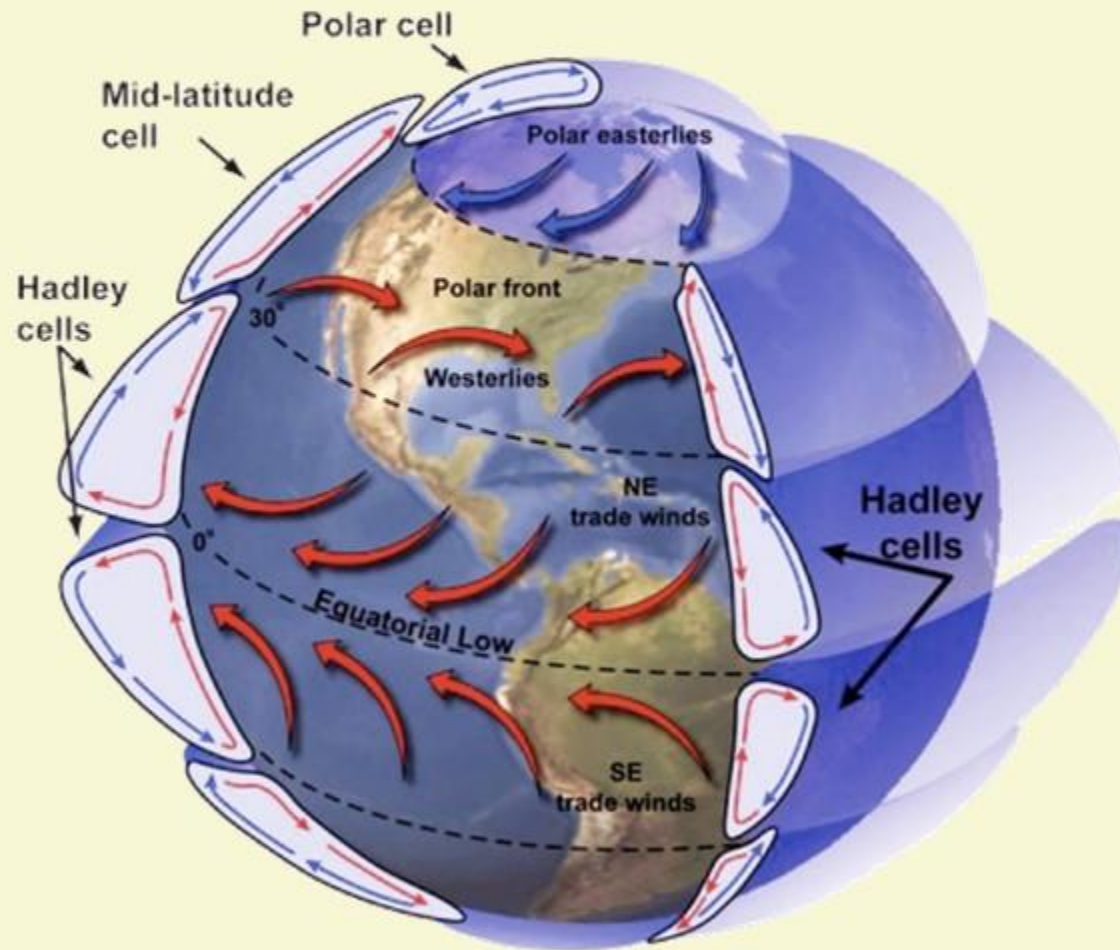
TROPICS & TRADE WINDS



- 1 There is **more or less regular inflow** of winds from subtropical high pressure belts to equatorial low pressure belt.
- 2 The tropical winds have north-easterly direction in the northern hemisphere while they are south-easterly in the southern hemisphere.
- 3 These winds are **called trade winds** because of the fact that they helped the **sea merchants in sailing their ships** as direction of trade winds remains more or less constant and regular.

PERMANENT OR PLANETARY WINDS

TROPICS & TRADE WINDS



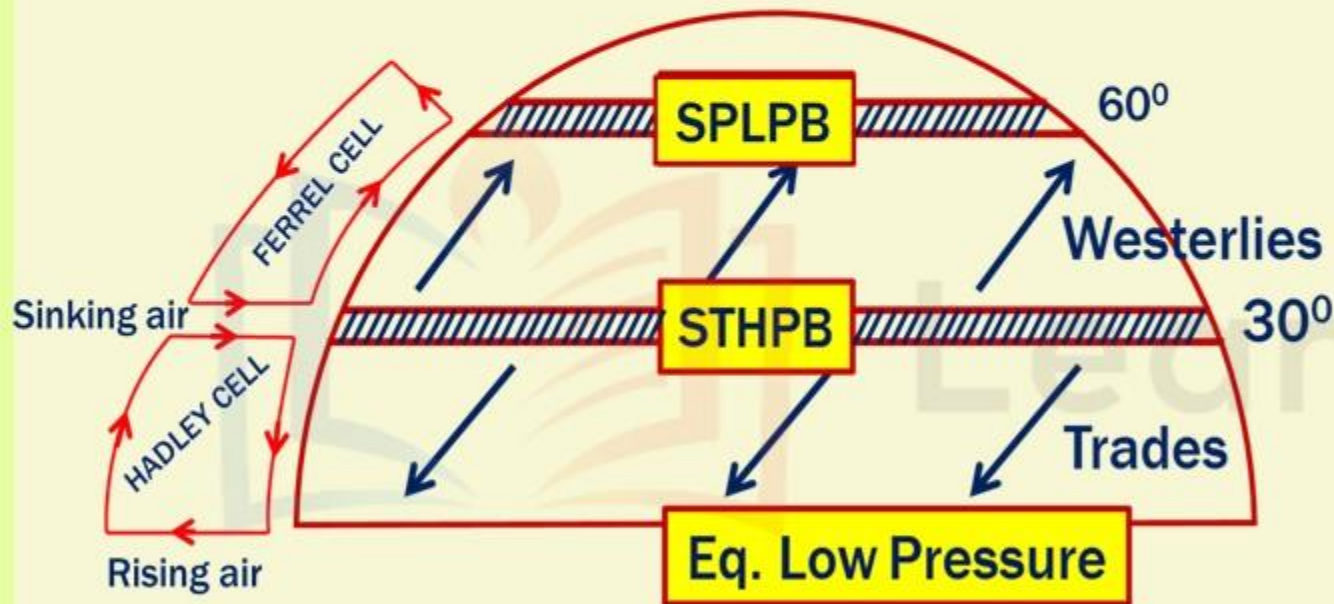
4 The poleward parts of the trade winds are dry because of strong subsidence of air currents from above.

5 Because of dominance of anticyclone conditions towards poleward parts there is strong atmospheric stability, strong inversion of temperature and clear sky.

6 On the other hand, the equatorward parts of the trade winds are humid because they are characterized by atmospheric instability and much precipitation as the trade winds while blowing over the oceans pick up moisture.

PERMANENT OR PLANETARY WINDS

HORSE LATITUDES



- 1 The dynamically induced subtropical high pressure belt **extends between 30⁰-35⁰** latitudes in both the hemispheres.
- 2 Thus, this belt **separates two wind systems** viz. trade winds and westerlies.
- 3 This subtropical high pressure belt is the source for the origin of trade winds (blowing towards equatorial low pressure belt) and westerlies (blowing towards subpolar low pressure belt)
- 4 The air after being heated near the equator ascends and after **blowing in opposite direction to the surface trade winds** descends in the latitudinal zone of 30⁰-35⁰

PERMANENT OR PLANETARY WINDS

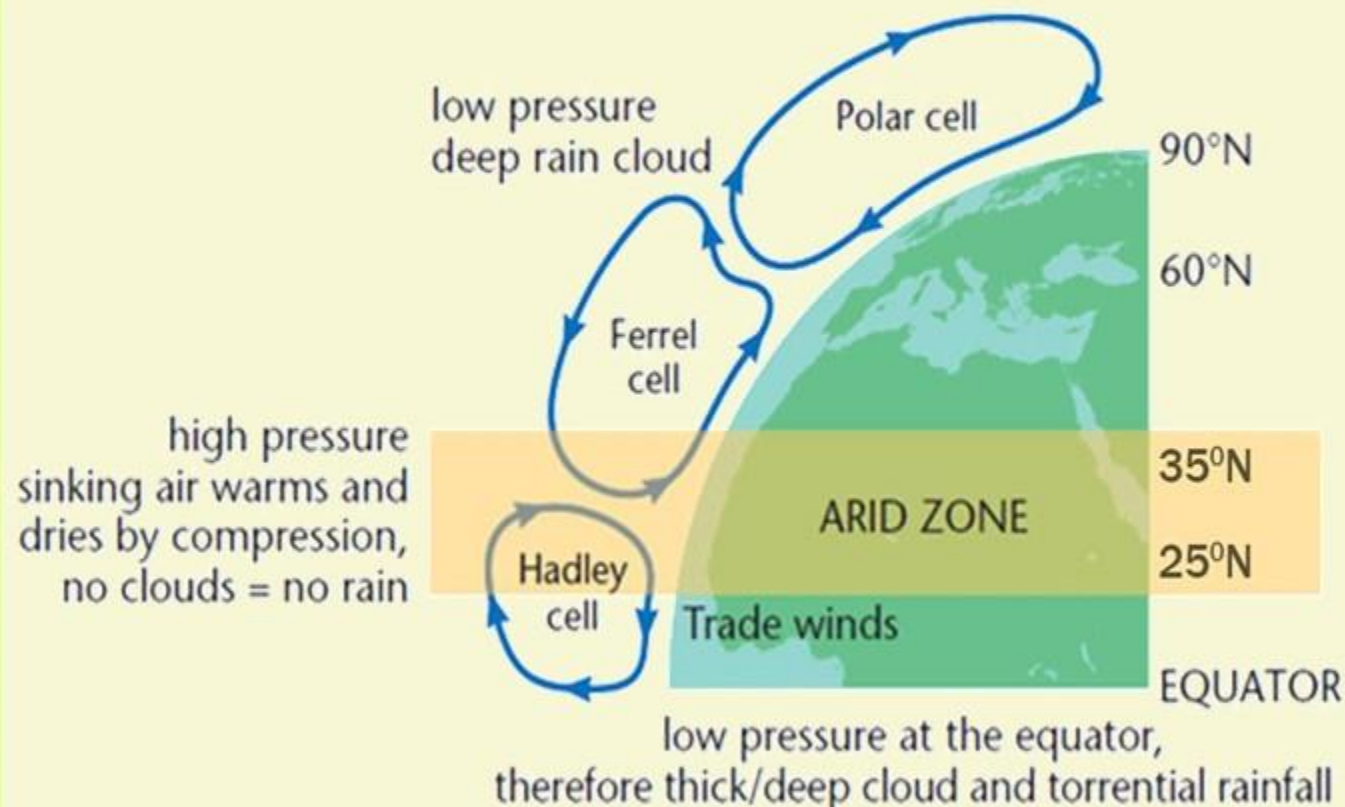
HORSE LATITUDES

5 This **descent of winds** from above causes high pressure on the surface which in turn causes **anticyclonic conditions**.

6 This is why the anticyclonic conditions cause atmospheric stability, dry condition and very weak air circulation.

7 Anticyclones are produced due to subsidence of air in the horse latitudes. These anticyclones are known as 'subtropical highs' or 'subtropical anticyclones'.

8 This is why hot and dry tropical deserts are found in western parts of the continents within the latitudinal zones of 20° - 30° in both the hemispheres



ATMOSPHERIC PRESSURE

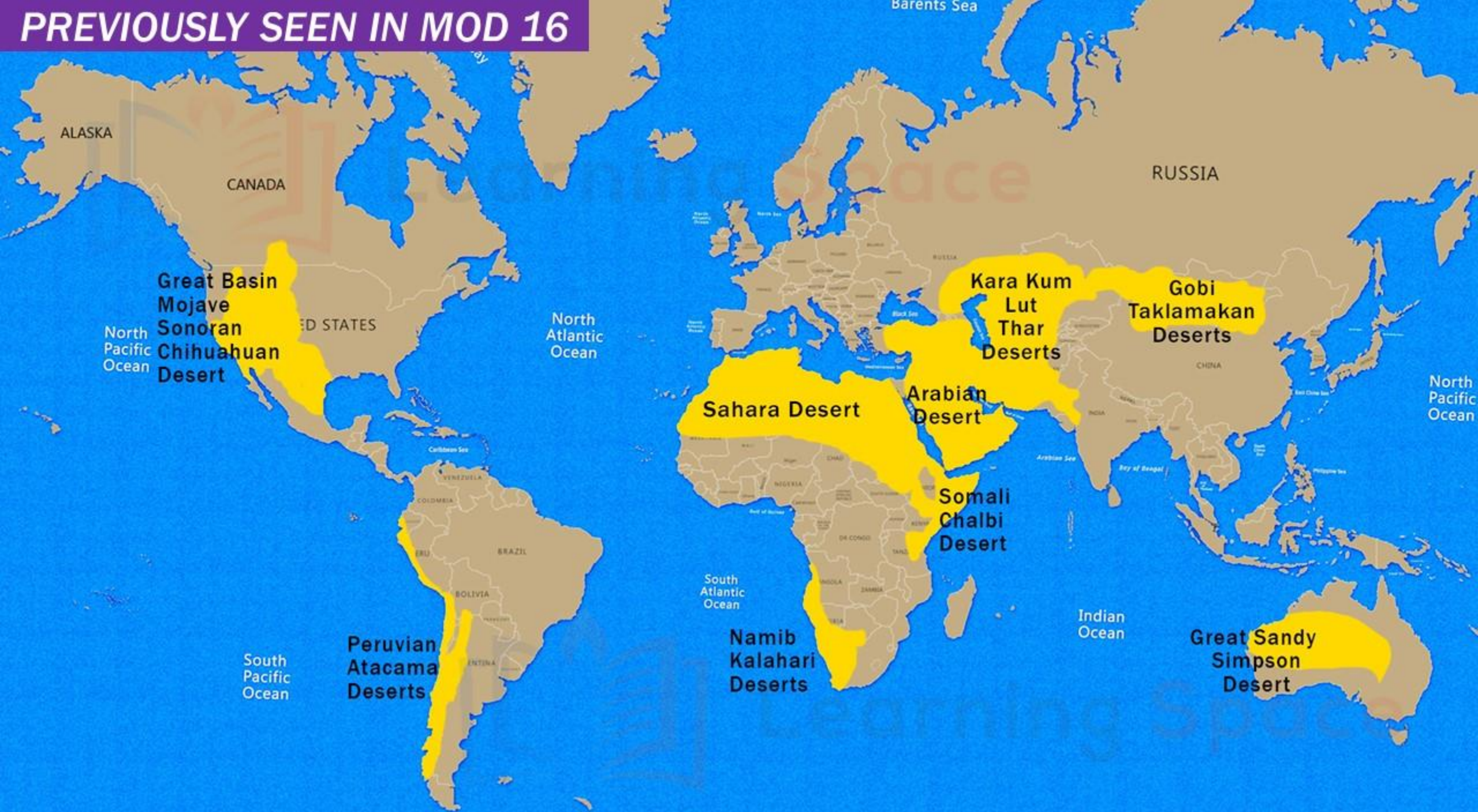
UPSC 2013 GS-1 QUESTION

Major hot deserts in northern hemisphere are located between 20-30 degree north and on the western side of the continents. Why? (10 Marks)

HINTS

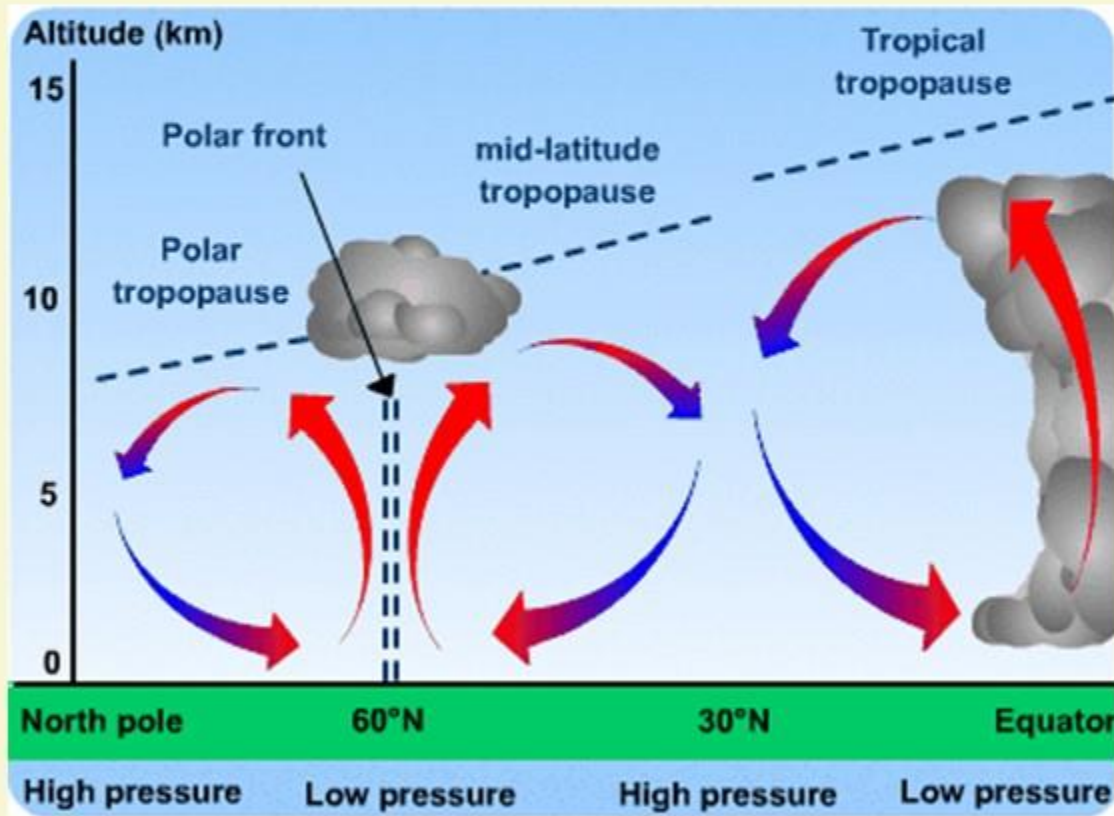
- 1 Offshore trade winds in the region
- 2 Prevailing anticyclonic conditions- Areas between 20-30 degree latitudes on western margins of continents are the regions of descending air
- 3 Presence of cold ocean currents along the western coast of continents- this stabilizes the air and prevents cloud formation and rainfall.
- 4 Leeward sides of mountains/Parallel mountain ranges

PREVIOUSLY SEEN IN MOD 16



PERMANENT OR PLANETARY WINDS

WESTERLIES



- 1 The permanent winds blowing from the subtropical high pressure belts (30° - 35°) to the subpolar low pressure belts (60° - 65°) in both the hemispheres is called westerlies
- 2 Towards the poleward parts there is convergence of cold and denser polar winds and warm and lighter westerlies.
- 3 In fact, a cyclonic front, called as polar front, is formed due to this convergence of two contrasting air masses and thus temperate cyclones are originated.
- 4 These cyclones move along with the westerlies in easterly direction.

PERMANENT OR PLANETARY WINDS

WESTERLIES



One of the observatory vessels of the NOAA plunging into the Roaring Forties

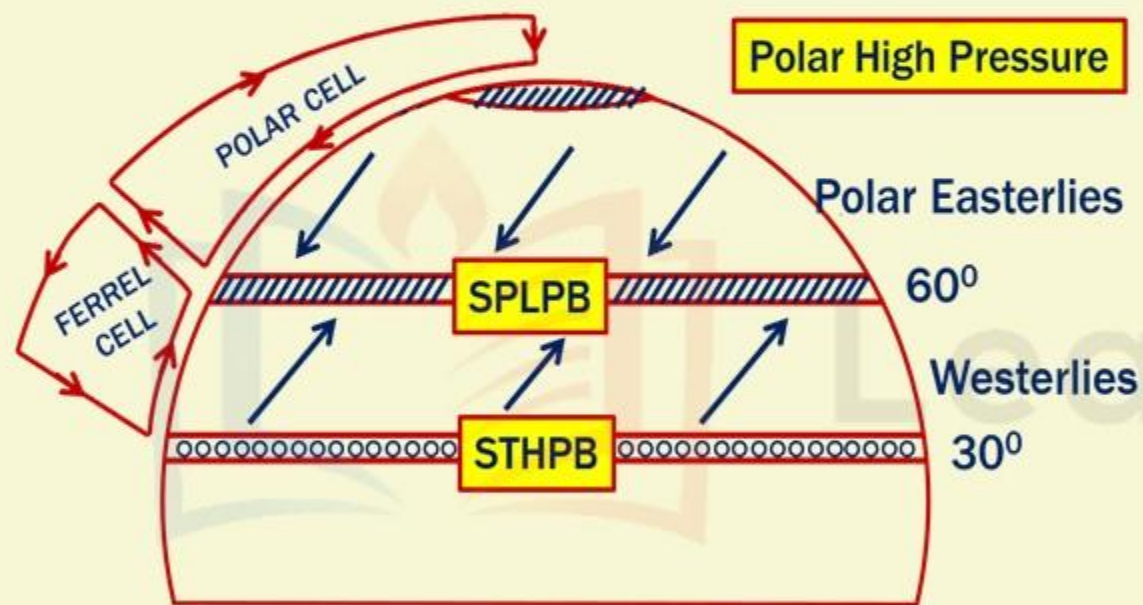
5 These westerlies bring much precipitation in the western parts of the continents (e.g. north-west European coast) because they pick up much moisture while passing over the vast stretches of the oceans.

6 The westerlies become more vigorous in the southern hemisphere because of lack of land and dominance of oceans.

7 The velocity of the westerlies becomes so great that they are called **Roaring Forties** between the latitudes of 40° - 50° S, **Furious Fifties** at 50° S latitude and **Shrieking Sixties** at 60° S latitude.

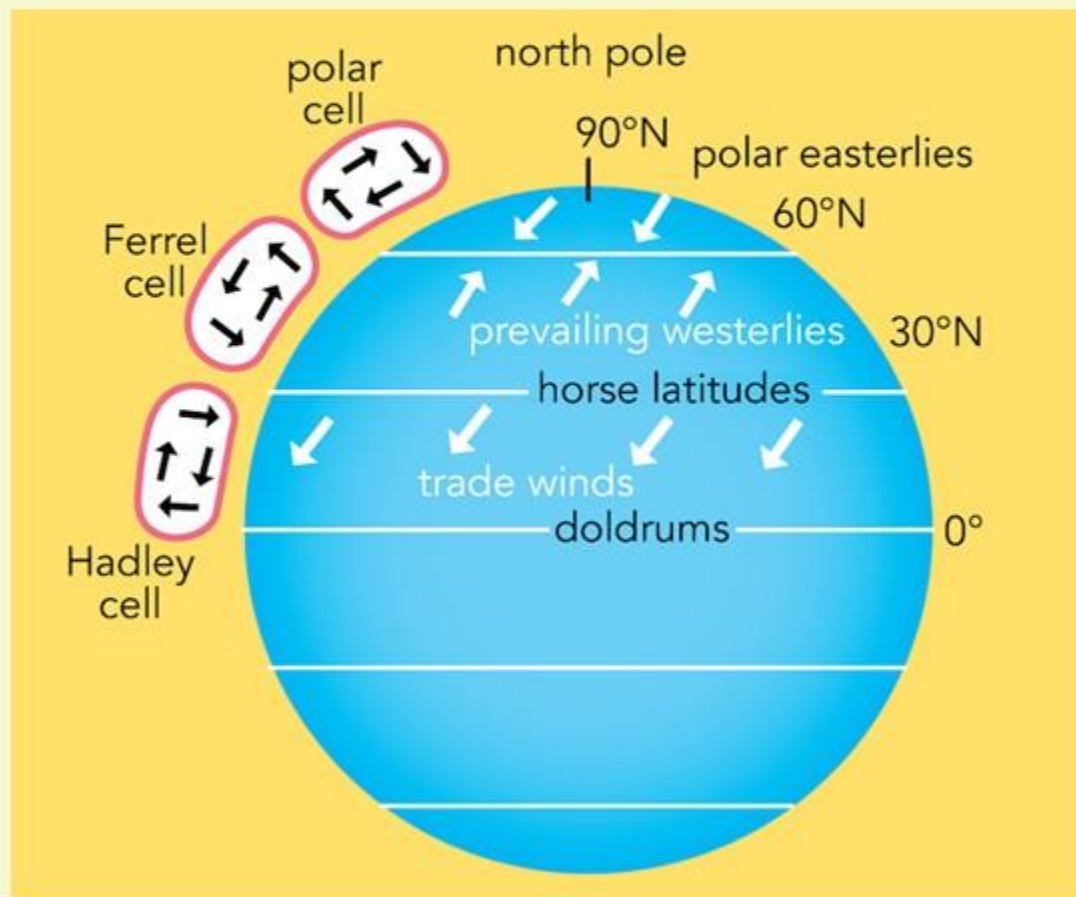
PERMANENT OR PLANETARY WINDS

POLAR WINDS



- 1 A low pressure belt, produced due to dynamic factor, lies within the latitudinal belt of 60° - 65° in both the hemispheres.
- 2 There is very high pressure over the poles because of exceedingly low temperature.
- 3 Thus, winds blow from polar high pressure to subpolar low pressure cells. These are called **polar easterlies**.
- 4 The **zone of polar winds shrinks** due to northward shifting of pressure belts at the time of the northern summer (summer solstice) in the northern hemisphere but it is extended up to 60° N latitude during the northern winter (winter solstice).

TRICELLULAR MERIDIONAL CIRCULATION OF THE ATMOSPHERE



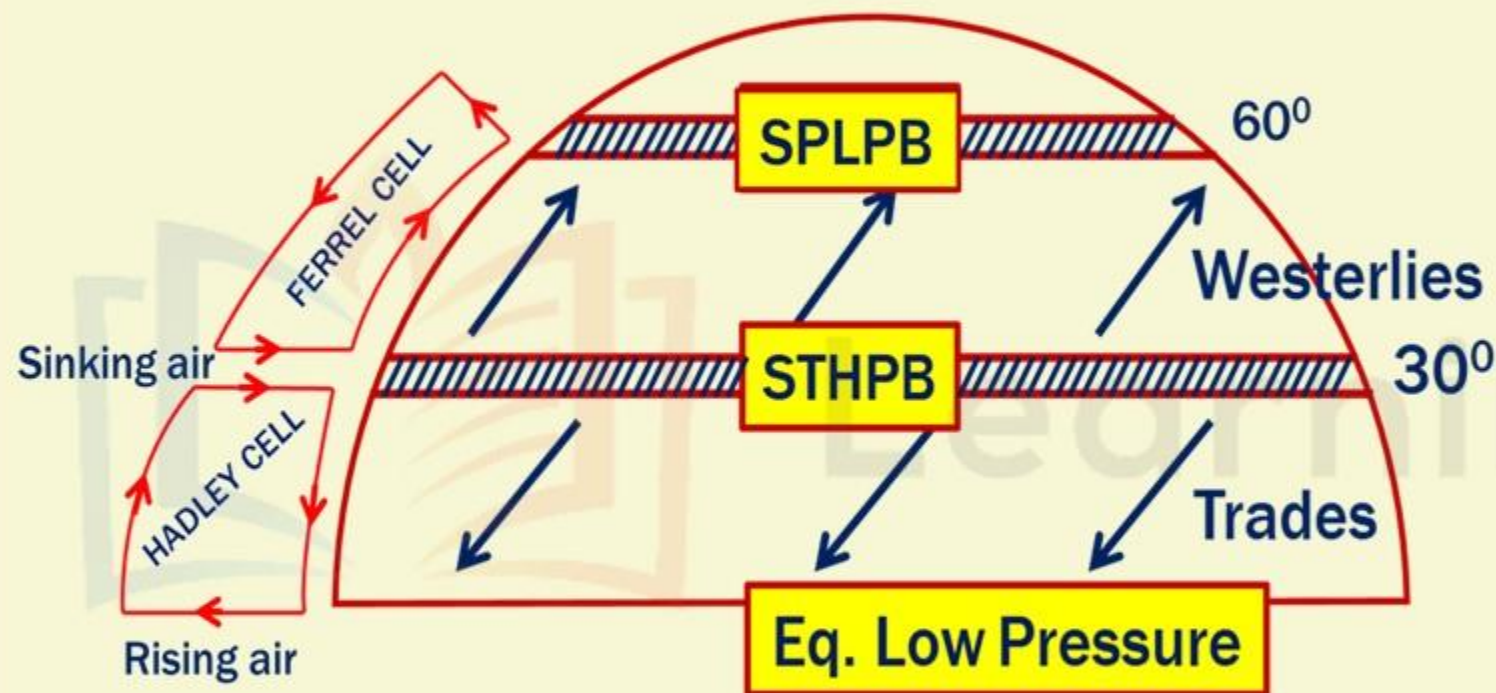
1 There exists a three-cell model of meridional circulation of the atmosphere popularly known as **tricellular meridional circulation of the atmosphere**.

2 Thus, each meridian has three cells of air circulation in the northern hemisphere

- Tropical cell or Hadley cell
- Mid-latitude cell or Ferrel cell
- Polar cell

TRICELLULAR MERIDIONAL CIRCULATION OF THE ATMOSPHERE

HADLEY CELL

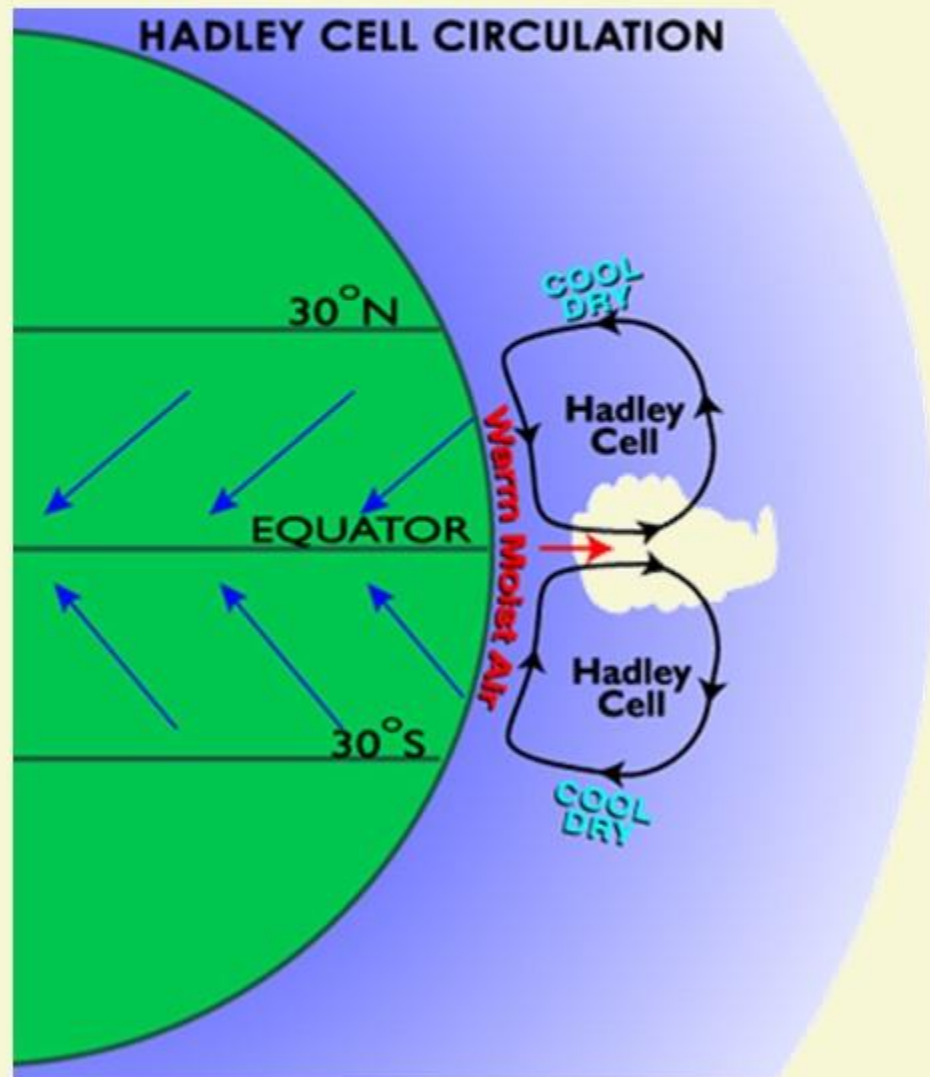


- 1 Tropical cell is also called as Hadley cell because G.Hadley first identified this thermally induced cell in both the hemispheres in the year 1735.

- 2 The winds after being heated due to very high temperature at the equator ascend upward.
- 3 These ascending warm, moist winds which after reaching the height of 8 to 12 kilometres in the troposphere over the equator diverge northward and southward or say poleward.
- 4 The surface winds in the name of trade winds blow from subtropical high pressure belts to equatorial low pressure belt in order to replace the ascending air at the equator.

TRICELLULAR MERIDIONAL CIRCULATION OF THE ATMOSPHERE

HADLEY CELL



5 The upper air moving in opposite direction to surface winds (trade winds) is called antitrades.

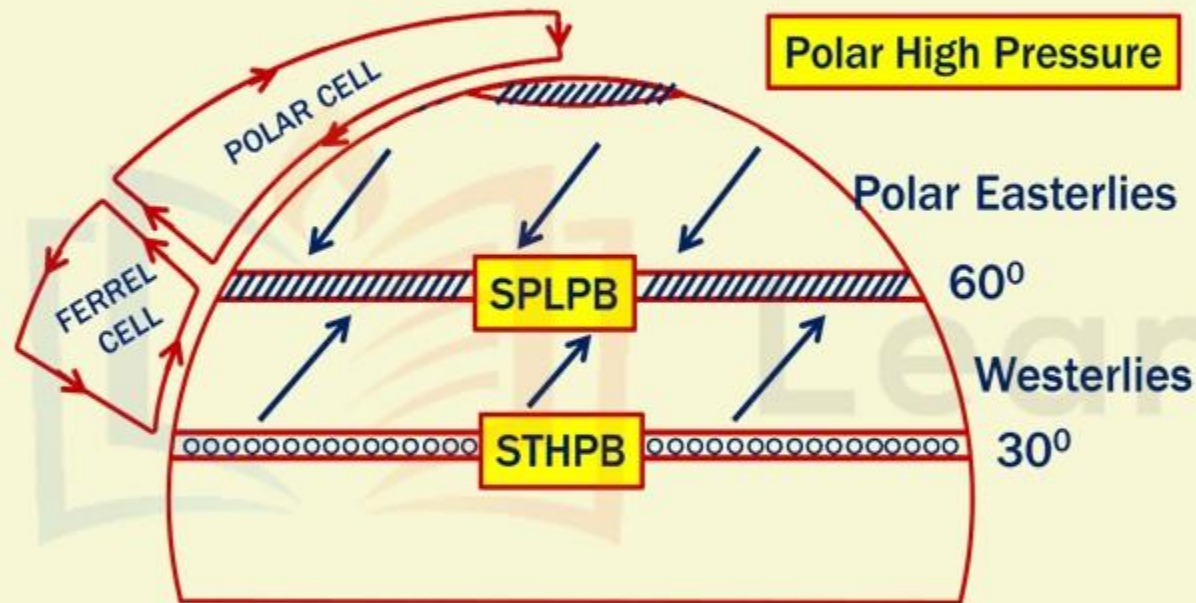
6 These upper air antitrades descend near 30° - 35° latitudes to cause subtropical high pressure belt.

7 These antitrades after descending near 30° - 35° latitudes again blow towards the equator where they are again heated and ascend.

8 Thus, one complete meridional cell of air circulation is formed. This is called tropical meridional cell which is located between the equator and 30° latitudes

TRICELLULAR MERIDIONAL CIRCULATION OF THE ATMOSPHERE

FERREL CELL

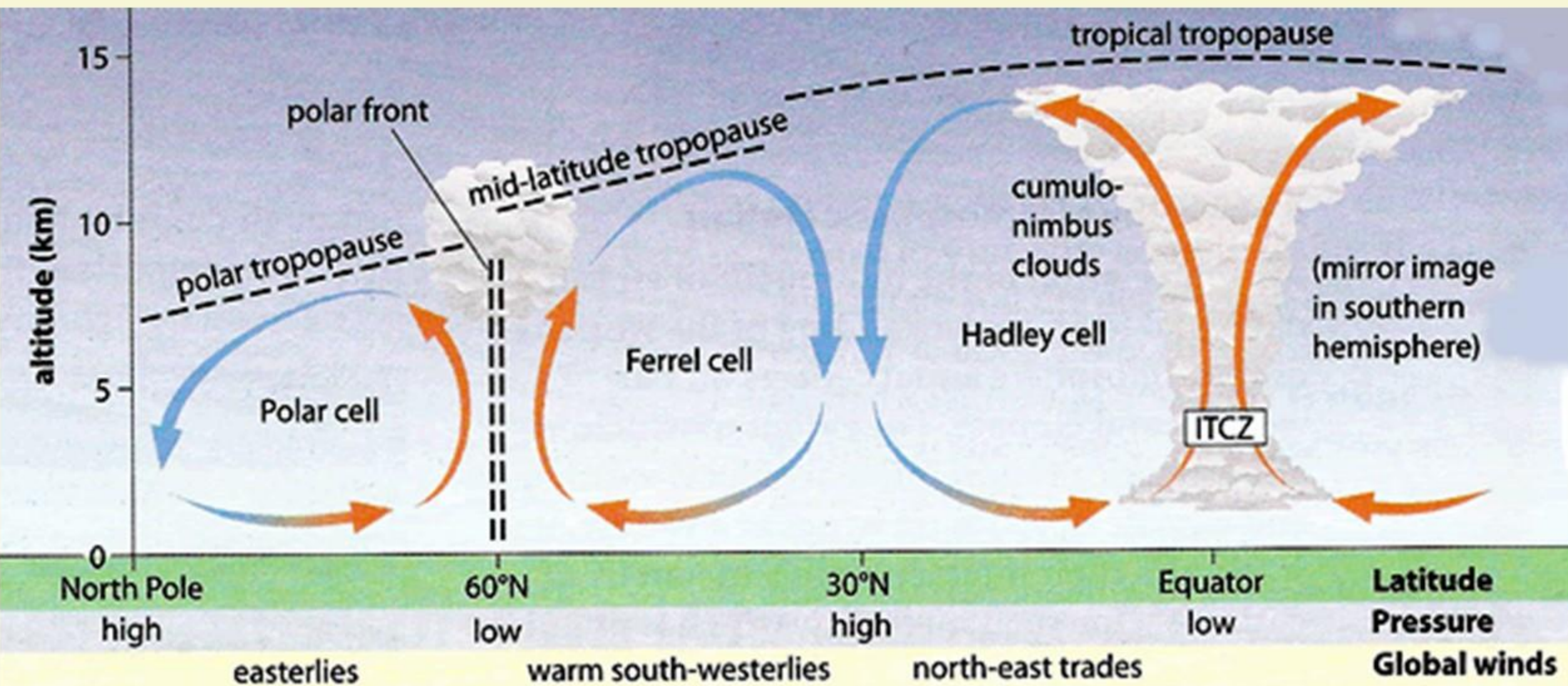


1 Surface winds, know as westerlies, blow from STHPB to SPLPB (60° - 65°).

2 The winds ascend near 60° - 65° latitudes because of the rotation of the Earth and after reaching the upper troposphere diverge in opposite directions (poleward and equatorward)

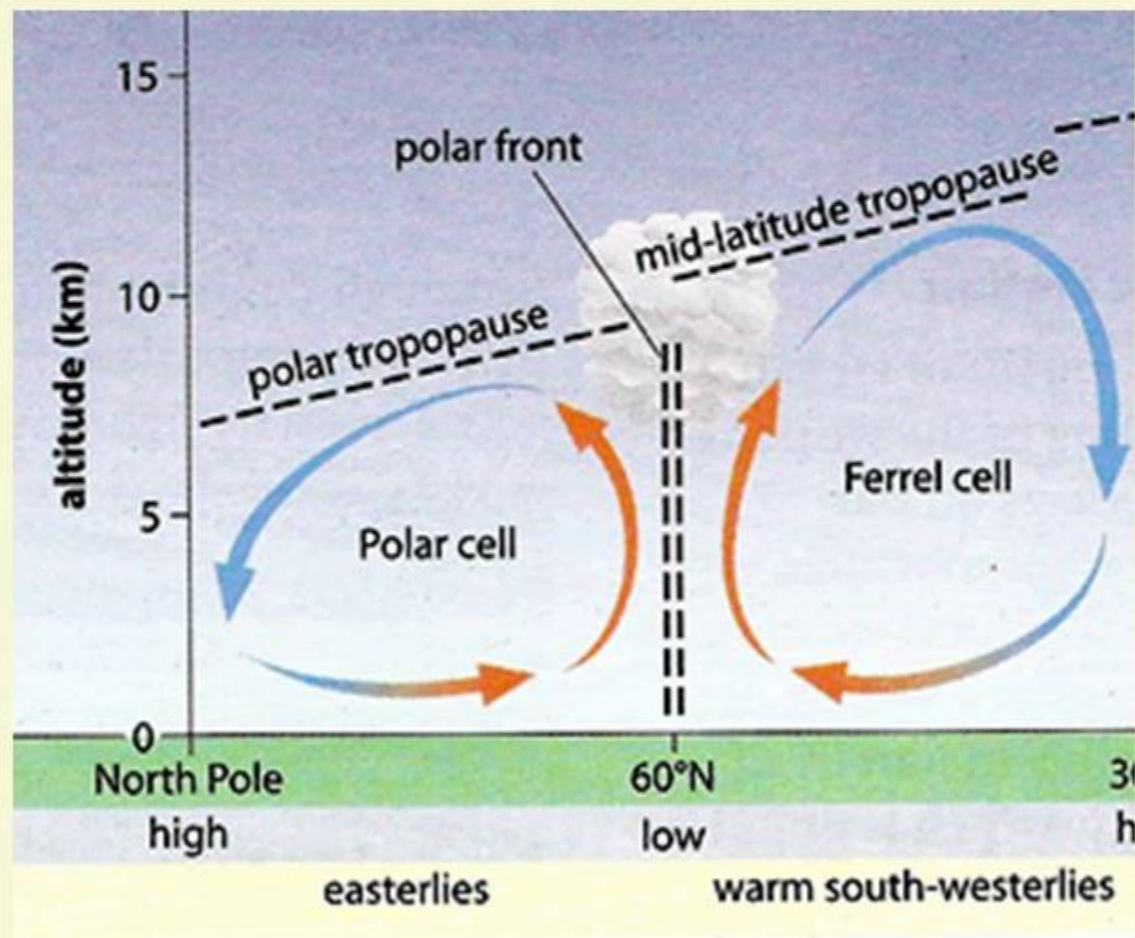
3 These winds (which diverge equatorward) again descend near horse latitudes (30° - 35° latitudes) to reinforce subtropical high pressure belt.

4 After descending these winds again blow poleward as surface westerlies and thus a complete cell is formed



TRICELLULAR MERIDIONAL CIRCULATION OF THE ATMOSPHERE

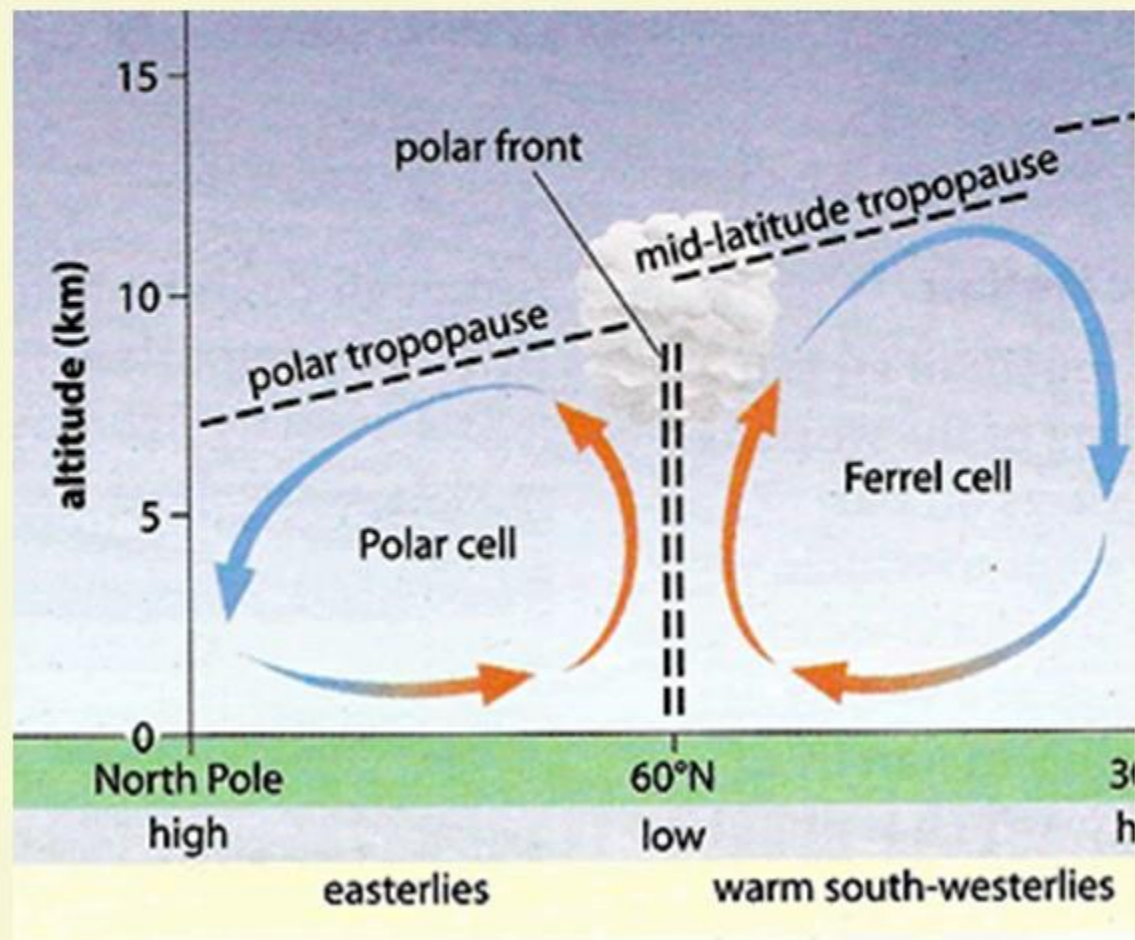
POLAR CELL



- 1 Polar cell involves the atmospheric circulation prevailing between 60° latitude and poles.
- 2 Cold winds, known as polar easterlies, blow from polar high pressure areas to sub-polar low pressure belt.
- 3 The general direction of surface polar winds becomes easterly (east to west) due to Coriolis force.

TRICELLULAR MERIDIONAL CIRCULATION OF THE ATMOSPHERE

POLAR CELL

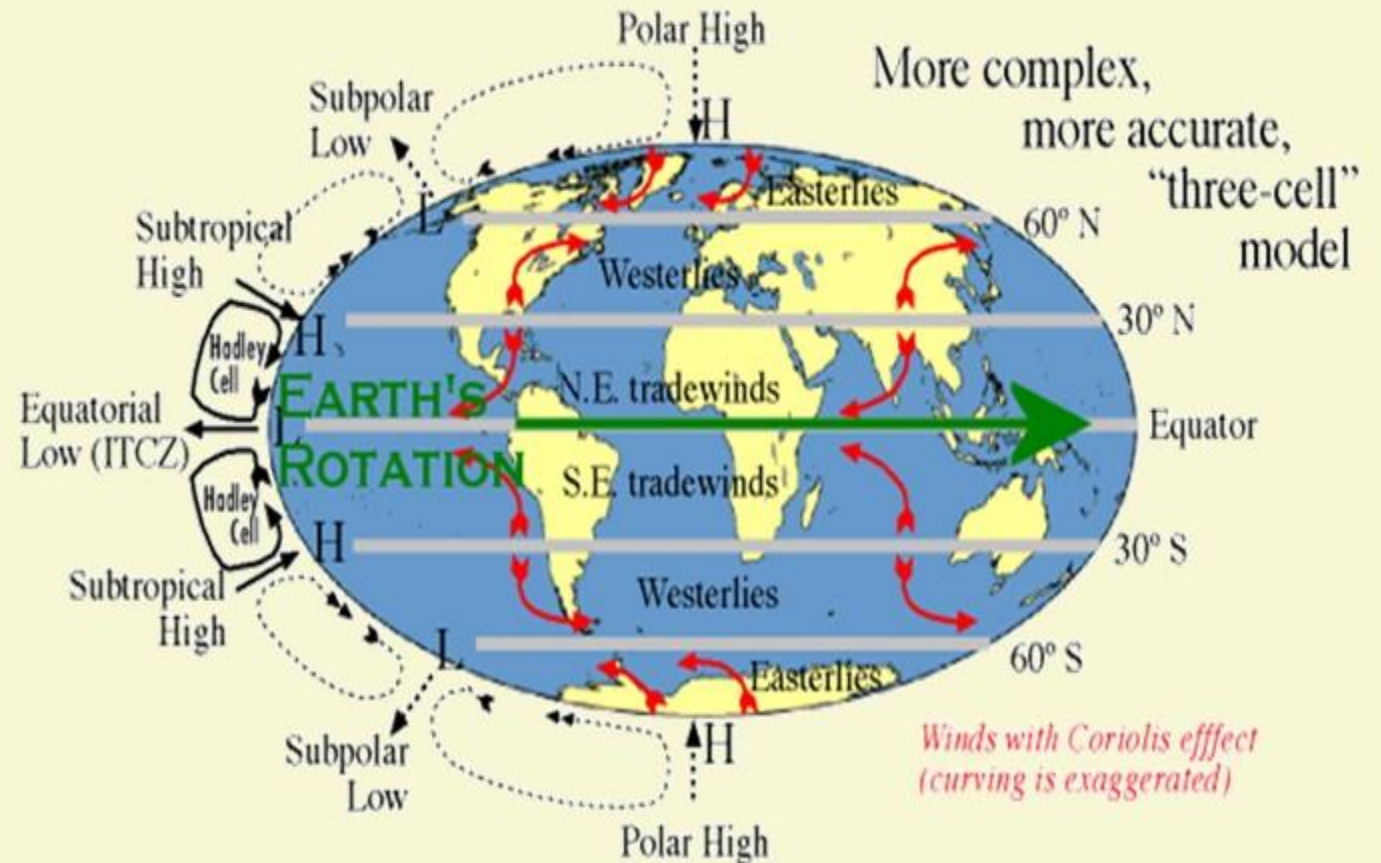
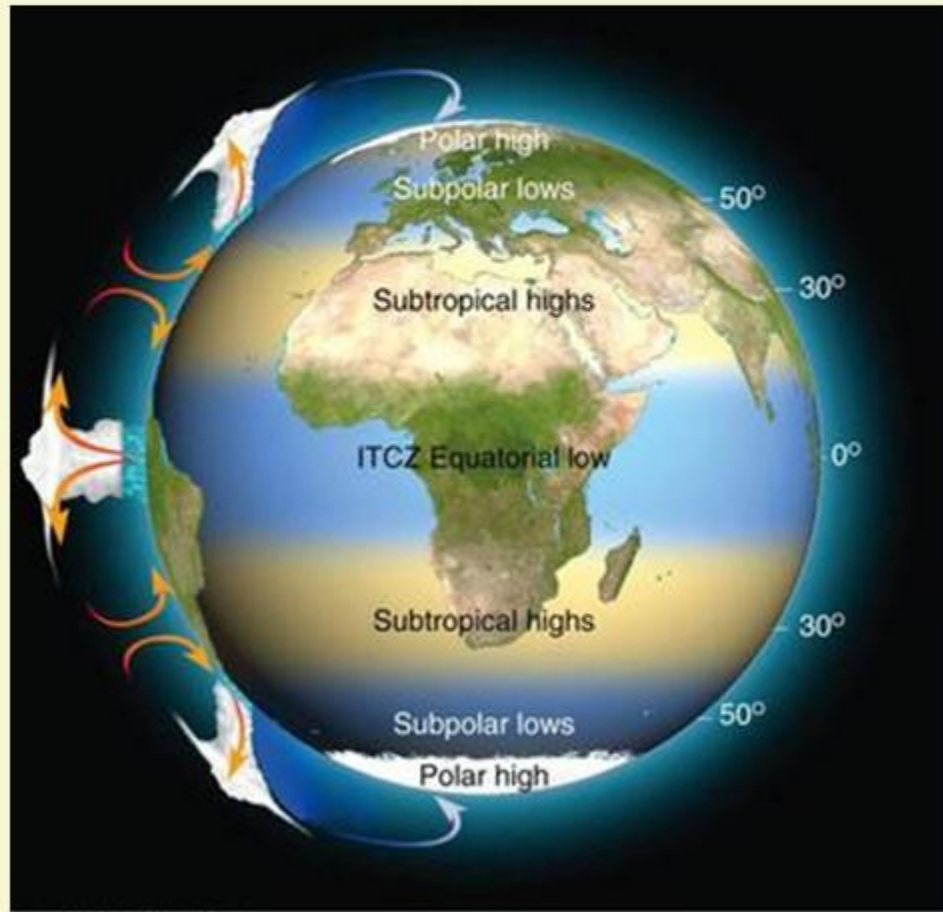


4 These polar winds converge with warm westerlies near 60° - 65° latitudes and form **polar front which becomes the source of temperature cyclones.**

5 The winds ascend due to the rotation of the Earth at the subpolar low pressure belt and after reaching troposphere they turn poleward and equatorward.

6 The poleward upper air descends at the poles and **reinforce the polar high pressure.** Thus, a complete polar cell is formed.

So to sum it up, we saw how the global pressure belts are distributed and how planetary winds originate because of the pressure belts.



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